WORKED SOLUTION

STEP II 2001, Question 1

Pure • Binomial Approximation

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q1

The key idea

The binomial series for $(1-x)^{1/2}$ converges quickly when x is small, so its first few terms give an excellent rational approximation. The skill is in choosing x: we want $1-x$ to be a square number times the number whose root we need, divided by a power of ten. Then the square root of $1-x$ is a simple multiple of the root we want. Since $99 = 9 \times 11$ and $9999 = 9 \times 1111$, the choices almost make themselves.

Solution

The quadratic approximation

By the binomial expansion, valid for $|x| < 1$,

$$\begin{aligned}(1-x)^{1/2} &= 1 + \tfrac{1}{2}(-x) + \frac{\frac{1}{2}\cdot\left(-\frac{1}{2}\right)}{2!}(-x)^2 + \frac{\frac{1}{2}\cdot\left(-\frac{1}{2}\right)\cdot\left(-\frac{3}{2}\right)}{3!}(-x)^3 + \cdots \\ &= 1 - \tfrac{1}{2}x - \tfrac{1}{8}x^2 - \tfrac{1}{16}x^3 - \cdots .\end{aligned}$$

So for small x,

$$\sqrt{1-x} \approx 1 - \frac{x}{2} - \frac{x^2}{8}.$$

We have also found the first neglected term, $-\frac{1}{16}x^3$, which we need for the errors. In fact every term after the first is negative (for $n \geq 1$ the product $\frac{1}{2}\cdot(-\frac{1}{2})\cdots(\frac{1}{2}-n+1)$ has $n-1$ negative factors, and $(-x)^n$ contributes $(-1)^n$, leaving one minus sign overall), so for $0 < x < 1$ the quadratic is an *overestimate*.

Part (i)

With $x = \frac{1}{100}$,

$$\sqrt{1-x} = \sqrt{\frac{99}{100}} = \frac{\sqrt{9\times 11}}{10} = \frac{3\sqrt{11}}{10}, \qquad \text{so} \qquad \sqrt{11} = \frac{10}{3}\sqrt{1-\tfrac{1}{100}}.$$

Using the quadratic approximation,

$$\sqrt{11} \approx \frac{10}{3}\left(1 - \frac{1}{200} - \frac{1}{80\,000}\right) = \frac{10}{3} \cdot \frac{80\,000 - 400 - 1}{80\,000}$$
$$= \frac{10}{3} \cdot \frac{79\,599}{80\,000} = \frac{79\,599}{24\,000},$$

as required.

Ask yourself The first neglected term is $-\frac{1}{16}x^3$. Is that the error in $\sqrt{11}$?

Not quite: that is the error in our approximation to $\sqrt{1-x}$, and we then multiplied by $\frac{10}{3}$. The error in $\sqrt{11}$ is therefore approximately

$$\frac{10}{3} \times \frac{1}{16}\left(\frac{1}{100}\right)^3 = \frac{10}{48 \times 10^6} = \frac{1}{4\,800\,000} \approx 2.08 \times 10^{-7}.$$

So, to 1 significant figure, the error is 2×10^{-7}, and $\frac{79\,599}{24\,000}$ is slightly larger than $\sqrt{11}$.

Part (ii)

We copy the trick of part (i). Since $1111 \times 9 = 9999 = 10^4 - 1$, take $x = \frac{1}{10\,000}$:

$$\sqrt{1-x} = \sqrt{\frac{9999}{10\,000}} = \frac{3\sqrt{1111}}{100}, \qquad \text{so} \qquad \sqrt{1111} = \frac{100}{3}\sqrt{1 - \tfrac{1}{10\,000}}.$$

Then

$$\sqrt{1111} \approx \frac{100}{3}\left(1 - \frac{1}{20\,000} - \frac{1}{8 \times 10^8}\right) = \frac{100}{3} \cdot \frac{800\,000\,000 - 40\,000 - 1}{8 \times 10^8}$$
$$= \frac{100}{3} \cdot \frac{799\,959\,999}{8 \times 10^8} = \frac{799\,959\,999}{24\,000\,000} = \frac{266\,653\,333}{8\,000\,000},$$

where in the last step we cancelled a factor of 3 (the digit sum of 799 959 999 is 75, which is divisible by 3).

The error is approximately $\frac{100}{3}$ times the first neglected term:

$$\frac{100}{3} \times \frac{1}{16}\left(\frac{1}{10^4}\right)^3 = \frac{100}{48 \times 10^{12}} \approx 2.08 \times 10^{-12},$$

which is about 2×10^{-12}, as the question asks. So a suitable rational number is

$$\sqrt{1111} \approx \frac{266\,653\,333}{8\,000\,000} \left(= \frac{799\,959\,999}{24\,000\,000}\right).$$

Checks

- $79\,599/24\,000 = 3.316625$, while $\sqrt{11} = 3.31662479\ldots$; the difference is 2.096×10^{-7}, matching the estimate 2.08×10^{-7} and confirming that the approximation is too big.

- $266\,653\,333/8\,000\,000 = 33.331666625$, while $\sqrt{1111} = 33.33166662291\ldots$; we computed the difference exactly to high precision and found 2.0835×10^{-12}.
- The estimate improves in part (ii) because the next neglected term, $-\frac{5}{128}x^4$, is smaller relative to the x^3 term when x is smaller.

Common slips

- Forgetting to multiply the neglected term by $\frac{10}{3}$ (or $\frac{100}{3}$), giving an error ten or a hundred times too small.
- Sign errors in the x^2 coefficient: with $(-x)^2$ the term is $\frac{\frac{1}{2}\cdot(-\frac{1}{2})}{2}x^2 = -\frac{1}{8}x^2$, not $+\frac{1}{8}x^2$.
- Trying x values that make $1 - x$ equal to 1111 divided by a power of ten. Such values are not small, so the series converges far too slowly to help.

Beyond STEP

The claim that the error is "about the first neglected term" can be made precise with Taylor's theorem and the Lagrange form of the remainder, which gives an exact formula for the error with an unknown point in $(0, x)$. From it we can get firm upper and lower bounds, not just estimates. Making these error bounds rigorous is a standard part of a first course in analysis; see Abbott, *Understanding Analysis*.

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